

MLPERF STORAGE V3.0 · SCALE-OUT AND PORTABILITY

AIStore scale-out and cloud portability

Validated OCI scaling across 3-, 6-, and 12-node deployments, plus three-cloud UNet3D portability.

3.97×

UNet3D training I/O scaling from 3 to 12 storage nodes

3.99×

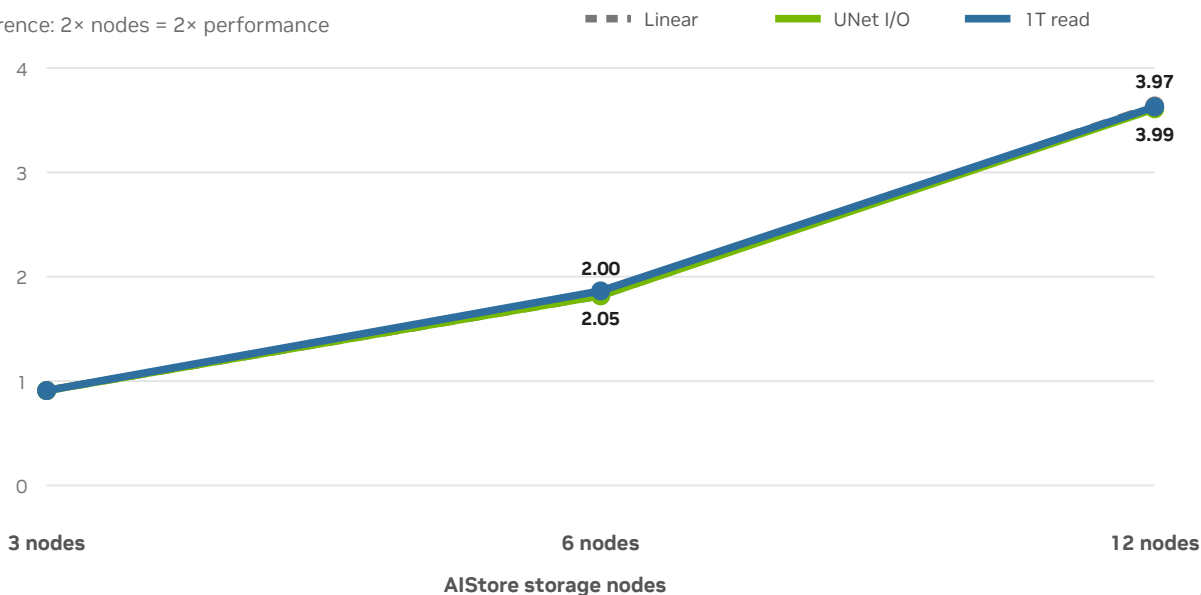
Llama 3 1T checkpoint recovery-read scaling from 3 to 12 nodes

97.7%

12-node 1T checkpoint read throughput versus nominal storage NIC line rate

Validated OCI performance tracks linear scaling

Reference: 2× nodes = 2× performance



Training

UNet3D performs 3D medical-image segmentation; RetinaNet performs object detection. The benchmark measures how quickly AIStore supplies training samples.

Checkpointing

Llama 3 1T emulates saving and recovering the state of a one-trillion-parameter model. The benchmark measures checkpoint write and read bandwidth.

AU · Accelerator Utilization

The estimated share of time simulated accelerators compute instead of waiting for storage. Higher AU means storage keeps them fed.

Validated metrics come directly from the packaged MLPerf submission artifacts.

UNet3D training scales linearly while AU stays high

UNet3D is MLPerf’s 3D medical-image segmentation training workload. It emulates B200 accelerators reading samples from AIStore through S3. Client and accelerator counts double with each storage tier.

115.58 GiB/s

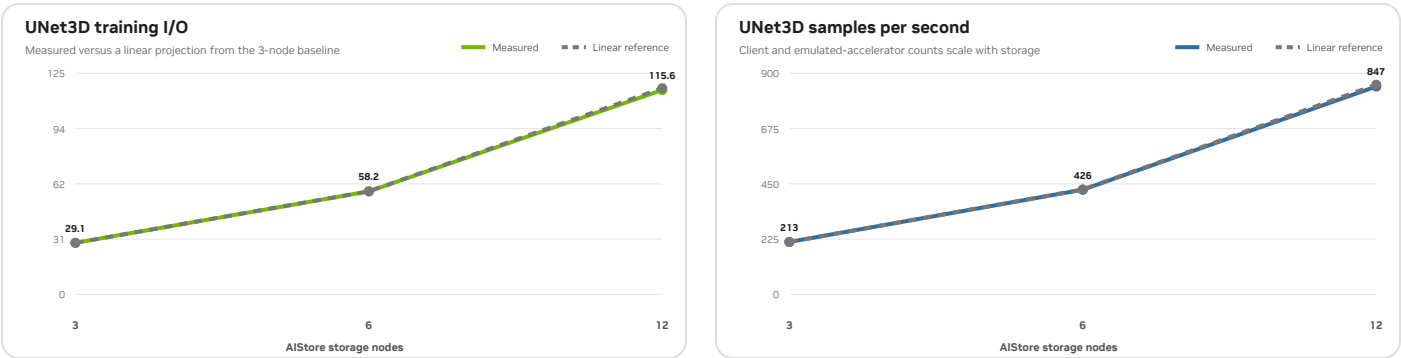
12-NODE TRAINING I/O

846.54

12-NODE SAMPLES/S

98.02%

12-NODE MEAN AU



Storage nodes	Clients	B200s	Mean AU	Samples/s	Training I/O	I/O scale
3	5	5	98.86%	213.47	29.15 GiB/s	1.00×
6	10	10	98.75%	426.44	58.22 GiB/s	2.00×
12	20	20	98.02%	846.54	115.58 GiB/s	3.97×

AU means Accelerator Utilization: the estimated percentage of benchmark time the simulated accelerators spend computing rather than waiting for storage. All three tiers exceed UNet3D’s 90% AU requirement; 12 nodes deliver 3.97× the 3-node I/O rate.

The dashed line is a projection from the refreshed measured three-node result, not a physical ceiling; small differences above or below it reflect variation between independent runs.

Llama 3 1T checkpointing operates near line rate

Llama 3 1T is the checkpoint workload for a one-trillion-parameter model. A checkpoint is a saved model-state snapshot used to resume training or recover after interruption.

133.27 GiB/s

12-NODE CHECKPOINT WRITE

136.54 GiB/s

12-NODE CHECKPOINT READ

3.99×

READ SCALE, 3 TO 12 NODES

1 · Save

Write ten finalized model-state checkpoints to AIStore.

**2 · Clear client cache**

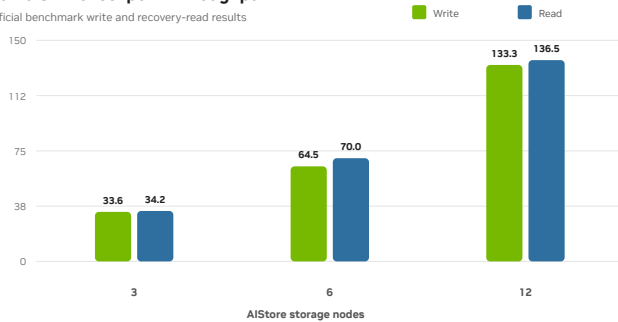
Remove cached data so recovery reads come from storage.

**3 · Recover**

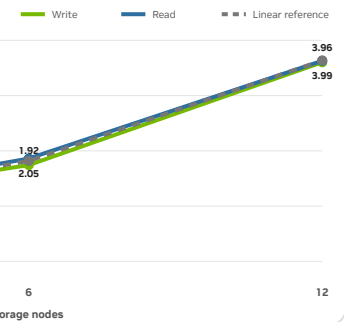
Read ten checkpoints back and measure aggregate GiB/s.

Llama 3 1T checkpoint throughput

Official benchmark write and recovery-read results

**1T checkpoint scaling**

Normalized to each 3-node measured result



Storage nodes	Clients	Write GiB/s	Read GiB/s	Nominal reference	Write vs nominal	Read vs nominal
3	8	33.6192	34.1959	34.9246	96.3%	97.9%
6	8	64.5050	70.0071	69.8492	92.3%	≈100%
12	16	133.2662	136.5435	139.6984	95.4%	97.7%

Result: the 12-node result reaches 95.4% of nominal write line rate and 97.7% of nominal read line rate. Its read result is 3.99× the 3-node baseline.

Each OCI storage node contributes an advertised 100 Gb/s interface, converted to a nominal 11.6415 GiB/s reference. The official six-node read score averages ten per-checkpoint rates; total bytes divided by mean elapsed time is 69.40 GiB/s, or 99.4% of nominal. The dashed scaling line is a baseline-derived reference, not a physical ceiling.

RetinaNet training and cloud portability

RetinaNet is MLPerf's object-detection training workload. Each image object is 322,957 bytes—about 323 kB (315 KiB). Each collective training step issues thousands of individual S3 GETs over HTTP/TCP, so request handling, object lookup, and protocol processing consume a larger share of time than payload transfer. Doubling from three to six storage nodes therefore delivers 1.85× I/O instead of an exact 2×. Mean AU remains above the required 85%: 94.06% at three nodes and 89.25% at six nodes.

1.85×

RETINANET I/O SCALING

89.25%

6-NODE RETINANET MEAN AU

3 clouds

VALIDATED S3 TRAINING PATH

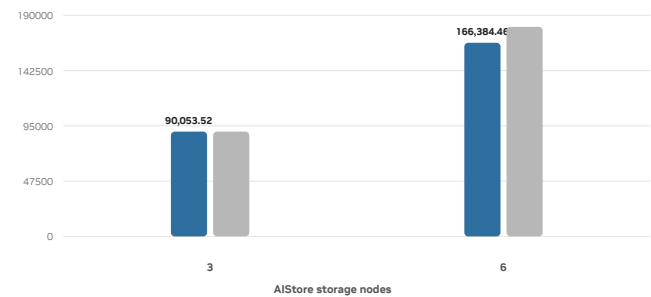
RetinaNet training I/O

Measured versus the linear 2× reference



RetinaNet samples per second

Measured versus the linear 2× reference



OCI · 3-node UNet3D

29.15

GiB/s

213.47

samples/s

98.86%

mean AU

AIStore storage: 3 × BM.DenseIO.E5.128

Benchmark clients: 5 ×

BM.DenseIO.E5.128 · 5 emulated B200s

AWS · 3-node UNet3D

46.41

GiB/s

339.93

samples/s

98.38%

mean AU

AIStore storage: 3 × i8ge.24xlarge

Benchmark clients: 4 × m8gn.24xlarge · 8

emulated B200s

GCP · 3-node UNet3D

46.15

GiB/s

338.04

samples/s

97.88%

mean AU

AIStore storage: 3 × z3-highmem-176-standardlssd

Benchmark clients: 8 × n2d-standard-96 · 8 emulated B200s

Interpretation boundary: OCI, AWS, and GCP three-node UNet3D results are not like-for-like cloud comparisons. Captured storage/client networking is 100/100 Gb/s on OCI, 150/300 Gb/s on AWS, and 200/100 Gb/s on GCP; shapes, client counts, datasets, rank counts, and workload tuning also differ. The GCP configuration was not tuned to saturate its aggregate nominal storage interfaces.